

2.2 Algorithm

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 02 of 13

★ Golden

2.2 Algorithm ★ Golden

LEARN IT

An **algorithm** is a step-by-step logical procedure used to solve a problem. It is **conceptual** in nature (it is an idea, a plan) and completely **independent of any programming language**. The same algorithm can be written later in Python, C++ or Java.

A good algorithm has:

- **Input** — the data it starts with.
- **Clear steps** — each step has only one meaning.
- **A definite end** — it must finish after a limited number of steps.
- **Output** — the result it produces.

WORKED EXAMPLE

Algorithm to find the larger of two numbers

Step 1: Start

Step 2: Read A and B

Step 3: If $A > B$, print "A is larger"

Step 4: Else print "B is larger or equal"

Step 5: End

Notice: there is no Python or C++ here. Anyone — even a non-programmer — can follow these steps. That is what "language-independent" means.

2.2.1 Importance of Algorithms

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

REMEMBER IT

Remember the five reasons as **S-O-R-A-B**: **S**tructured, **O**ptimised, **R**eusable, **A**ccurate, **B**lueprint.

COMMON MISTAKE

Do not write "an algorithm is a program". A program is the algorithm *written in a programming language* so that a computer can run it. The algorithm itself is only the logic.

اردو میں سمجھیں

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔ یہ ایک تصور ہے اور کسی بھی پروگرامنگ زبان پر منحصر نہیں۔ الگورتھم کام کو منظم بناتے ہیں، وقت اور میموری بچاتے ہیں، دوبارہ استعمال ہو سکتے ہیں، غلطیاں کم کرتے ہیں اور کوڈ لکھنے سے پہلے نقشے کا کام دیتے ہیں۔

EXAM TIP

Very common short question: "Define algorithm and write its importance." Write the definition word for word — *a step-by-step logical procedure used to solve a problem; conceptual and independent of any programming language* — then list at least four of the five importance points. If asked to "write an algorithm", always begin with **Step 1: Start** and finish with **End**, and number every step.

CHECK YOURSELF

1. Why is an algorithm called "language-independent"?
2. Give two reasons why algorithms are important.
3. Write a 5-step algorithm to find the sum of two numbers.

Answers: 1. Because it describes only the logic; it is not tied to Python, C++ or any other language. 2. Any two: structured approach, optimise time/memory, reusable, accurate and consistent, blueprint before coding. 3. Step 1: Start; Step 2: Read A and B; Step 3: $SUM = A + B$; Step 4: Print SUM; Step 5: End.

Next lecture: 2.2.2 Algorithm vs Pseudocode